

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE PHYSICS

Paper 2

Tuesday 14 November 2023 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- the Physics Equations Sheet (enclosed).

Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- In all calculations, show clearly how you worked out your answer.

Information

- The maximum mark for this paper is 90.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	



Answer **all** questions in the spaces provided.

0 1

The Sun is a main sequence star.

The core of the Sun contains nuclei of two different isotopes of hydrogen (H).

0 1 . 1

Complete the following sentences about the isotopes of hydrogen in the Sun.

Choose answers from the box.

atoms

electrons

neutrons

protons

[2 marks]

The nuclei of the isotopes contain the same number of _____.

The nuclei of the isotopes contain a different number of _____.

The two hydrogen nuclei join together in a process that releases energy.

0 1 . 2

What is the name of this process?

[1 mark]

Tick (✓) **one** box.

Nuclear decay

☐

Nuclear fission

☐

Nuclear fusion

☐


0 1 . 3 Which element is created when two hydrogen nuclei join together?

[1 mark]

Tick (✓) **one** box.

Carbon ☐

Helium ☐

Nitrogen ☐

Oxygen ☐

0 1 . 4 Complete the sentence to describe the conditions in the core of the Sun.

[2 marks]

The conditions in the core of the Sun are high _____ and
high _____.

The Sun will leave the main sequence when it runs out of hydrogen in its core.

0 1 . 5 What is the minimum time the Sun will be a main sequence star?

[1 mark]

Tick (✓) **one** box.

Hundreds of years ☐

Thousands of years ☐

Millions of years ☐

Question 1 continues on the next page

Turn over ►



0 1 . 6

What is the next stage in the life cycle of the Sun after it leaves the main sequence?

[1 mark]

Tick (✓) **one** box.

Red giant

☐

Supernova

☐

White dwarf

☐

0 1 . 7

What are the **two** possible stages at the end of the life cycle of stars **much bigger** than the Sun?

[2 marks]

Tick (✓) **two** boxes.

Black dwarf

☐

Black hole

☐

Nebula

☐

Neutron star

☐

Protostar

☐

10



Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

Turn over ►



0 2

A car is stationary at a set of traffic lights.

0 2 . 1

When the traffic lights turn green the car accelerates.

It takes 12 seconds for the velocity of the car to increase by 30 m/s.

Calculate the acceleration of the car.

Use the Physics Equations Sheet.

[2 marks]

Acceleration = _____ m/s²

0 2 . 2

The mass of the car is 1500 kg.

Calculate the momentum of the car when the velocity of the car is 30 m/s.

Use the Physics Equations Sheet.

[2 marks]

Momentum = _____ kg m/s



Figure 1 shows marks left on a road surface by a lorry when the driver applied the brakes in an emergency.

Figure 1



The marks are made when a force removes some of the rubber from the tyres on the lorry.

The length of marks on the road are equal to the braking distance of the lorry.

0 2 . 3 What is the name of the force that acts between the surface of the road and the tyres on the lorry?

[1 mark]

Tick (✓) **one** box.

Friction

☐

Gravity

☐

Tension

☐

0 2 . 4 The speed of the lorry affects the length of the marks on the road.

Give **two other** factors that would affect the length of the marks.

[2 marks]

1 _____

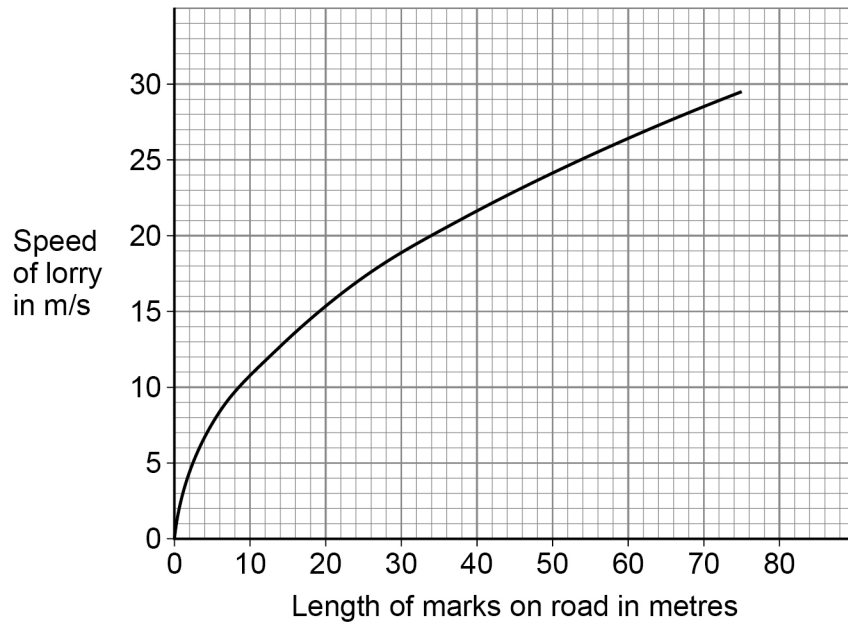
2 _____

Turn over ►



Figure 2 shows the relationship between the length of the marks on the road and the speed of the lorry.

Figure 2



0 2 . 5

Describe the relationship between the length of the marks on the road and the speed of the lorry.

Use **Figure 2**.

[2 marks]

The marks on the road in **Figure 1** on page 7 are 34 m long.

0 2 . 6 Determine the speed of the lorry when the brakes were applied.

Use **Figure 2**.

[1 mark]

Speed = _____ m/s

0 2 . 7 The work done by the brakes was 2 380 000 J.

Calculate the total force exerted by the brakes.

Use the Physics Equations Sheet.

Give the unit.

[4 marks]

Total force = _____ Unit _____

14

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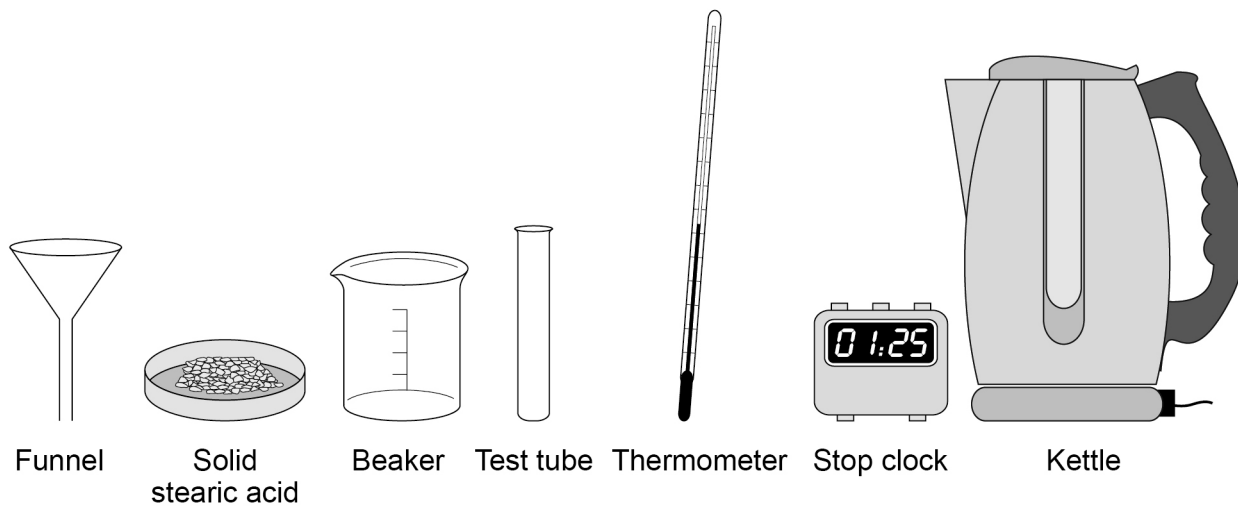
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A student investigated a cooling curve for stearic acid.

Figure 3 shows the equipment used.

Figure 3



0	3	.	1
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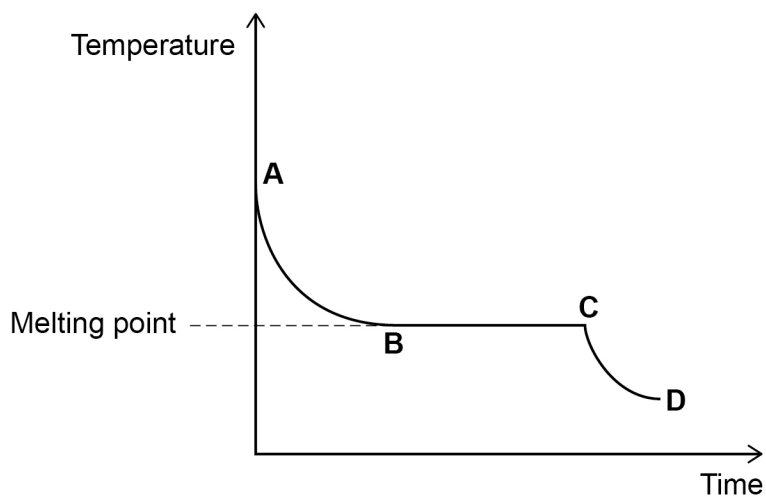
Describe a method to collect the results needed to draw a cooling curve for stearic acid.

[6 marks]

[illegible]

0 3 . 2 **Figure 4** shows a cooling curve for stearic acid as it cools.

Figure 4



Describe how the **movement** of the particles in the stearic acid changes between the different points labelled on the graph.

[3 marks]

From **A** to **B** _____

From **B** to **C** _____

From **C** to **D** _____



0	3	.	3
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It takes 3600 J of energy to melt a mass of 18 g of stearic acid without changing the temperature.

Calculate the specific latent heat of fusion of stearic acid.

Use the Physics Equations Sheet.

[3 marks]

Specific latent heat of fusion = _____ J/kg

12



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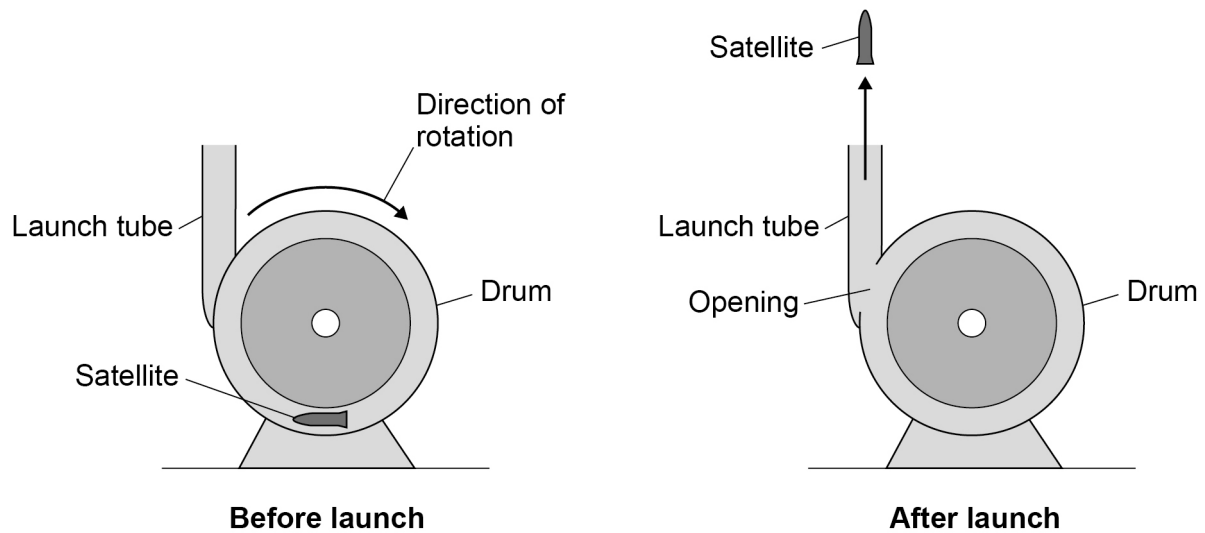


0 4

Scientists are developing a rotating drum system to launch satellites from Earth into space without needing a rocket.

Figure 5 shows the system.

Figure 5



A satellite is placed inside a drum. The drum rotates until the satellite is moving fast enough to be launched out of the opening.

The satellite is then launched upwards out of the launch tube.

The satellite does not burn any fuel after being launched.

0 4 . 1

Describe the energy transfers that take place after the launch in **Figure 5**.

[3 marks]



0 4 . 2

Name the **two** forces that act on the satellite after the launch in **Figure 5**.**[2 marks]**

1 _____

2 _____

0 4 . 3

Explain why these two forces cause the velocity of the satellite to decrease.

[3 marks]

0 4 . 4

The satellite has a mass of 200 kg.

The kinetic energy of the satellite when it is launched is 625 MJ.

Calculate the velocity of the satellite when it is launched.

Use the Physics Equations Sheet.

[4 marks]

Velocity = _____ m/s

Question 4 continues on the next page**Turn over ►**

0 4 . 5

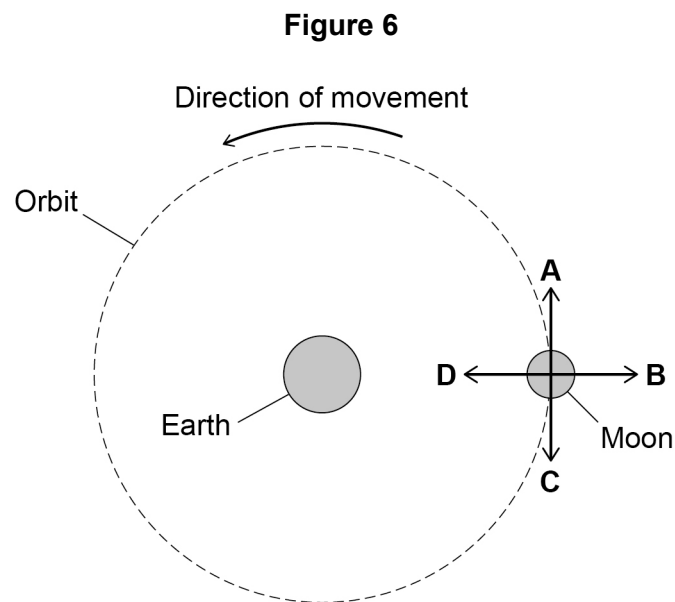
The rotating drum system is designed to launch the satellite into a low orbit.

Explain how the rotating drum system would need to be changed to launch the satellite into a higher orbit.

[3 marks]

The Moon is the only natural satellite of the Earth.

Figure 6 shows the orbit of the Moon around the Earth.



The Moon is accelerating.

0 4 . 6

Which of the arrows **A**, **B**, **C** or **D** on **Figure 6** shows the direction of the Moon's acceleration?

[1 mark]

Tick (✓) **one** box.

A ☐

B ☐

C ☐

D ☐

0 4 . 7

The Moon is travelling at a constant speed.

Explain how the Moon can be accelerating and travelling at a constant speed.

[3 marks]

19

Turn over for the next question

Turn over ►



0 5

Electricity is distributed from power stations along high-voltage transmission cables.

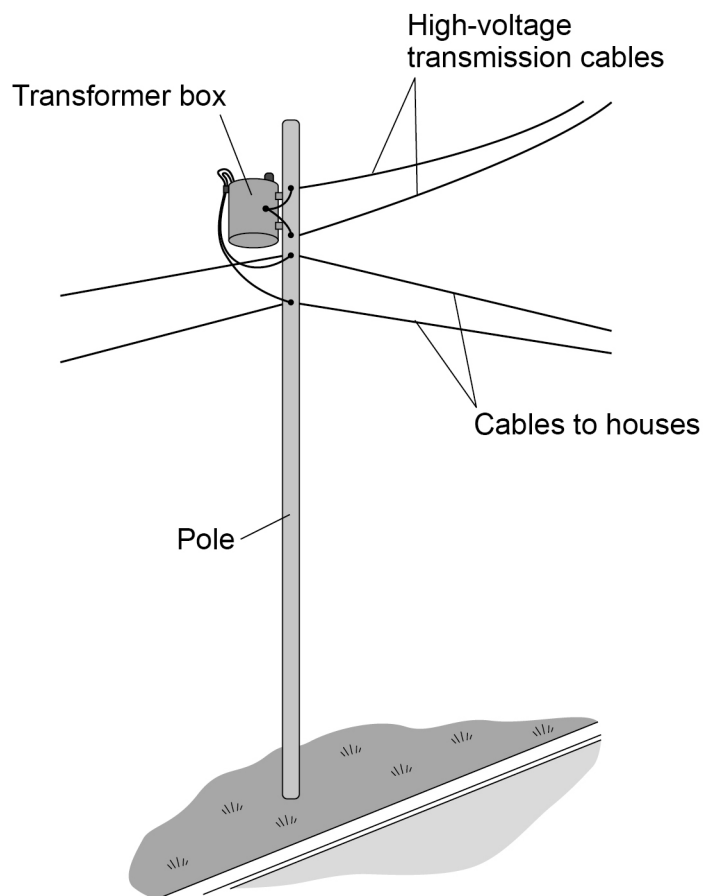
0 5**1**

Explain why the electricity is distributed at a very high voltage.

[3 marks]

Figure 7 shows high-voltage transmission cables connected to a transformer box.

Figure 7



The transformer box contains a step-down transformer that reduces the voltage.

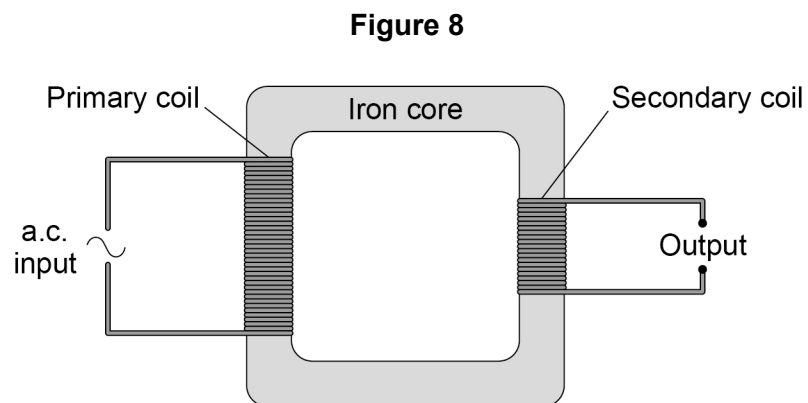
Cables then transmit the lower-voltage electricity to houses.

0 5 . 2

Give the reason why the step-down transformer is used to reduce the voltage.

[1 mark]

Figure 8 shows the step-down transformer in the transformer box.



Question 5 continues on the next page

Turn over ►



0 5 . 3

In the transformer the ratio of the number of turns on the two coils is:

$$\frac{\text{number of turns on the primary coil}}{\text{number of turns on the secondary coil}} = 30$$

The a.c. input is 7.2 kV.

Calculate the output voltage.

Use the Physics Equations Sheet.

[3 marks]

Output voltage = _____ V

0 5 . 4

In some countries the cables that supply electricity to houses are buried under the ground.

Suggest **one** advantage and **one** disadvantage of having underground cables.

[2 marks]

Advantage _____

Disadvantage _____



0 6

A student investigated how the current in a filament lamp is affected by the potential difference across the filament lamp.

0 6 . 1

The student used the following equipment:

- cell
- switch
- variable resistor
- filament lamp
- ammeter
- voltmeter.

Draw a circuit diagram for a circuit the student could use.

[3 marks]

Question 6 continues on the next page

Turn over ►

0 6 . 2 Table 1 shows the results from the investigation.

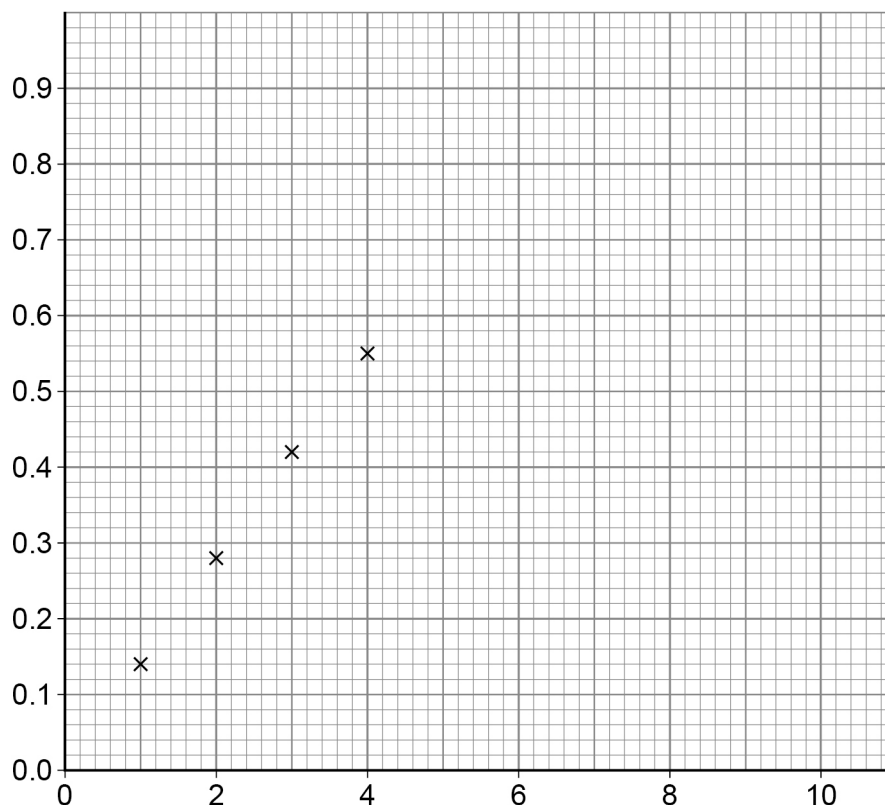
Table 1

Potential difference in volts	Current in amps
1.0	0.14
2.0	0.28
3.0	0.42
4.0	0.55
5.0	0.65
6.0	0.73
7.0	0.79
8.0	0.83
9.0	0.85



Figure 9 is an incomplete graph for the data in **Table 1**.

Figure 9



Complete **Figure 9**.

You should:

- label both axes
- plot the remaining data from **Table 1**
- draw a line of best fit.

[4 marks]

0 6 . 3

The filament lamp is designed to get hot, so does **not** need to be disconnected to prevent heating.

Suggest why it is still a good idea to switch the lamp off between readings.

[1 mark]

Question 6 continues on the next page

Turn over ►



0 6 . 4

Explain why the resistance of a filament lamp increases with temperature.

[2 marks]

0 6 . 5

A high-power filament lamp contains a filament made of the metal tungsten.

The average power of the filament lamp is 520 W.

When the lamp was switched on it took 1.1 s for the temperature of the filament to increase by 2200 °C.

specific heat capacity of tungsten = 130 J/kg °C

Calculate the mass of the tungsten filament.

Use the Physics Equations Sheet.

Give your answer in grams.

[6 marks]

Mass = _____ g

16

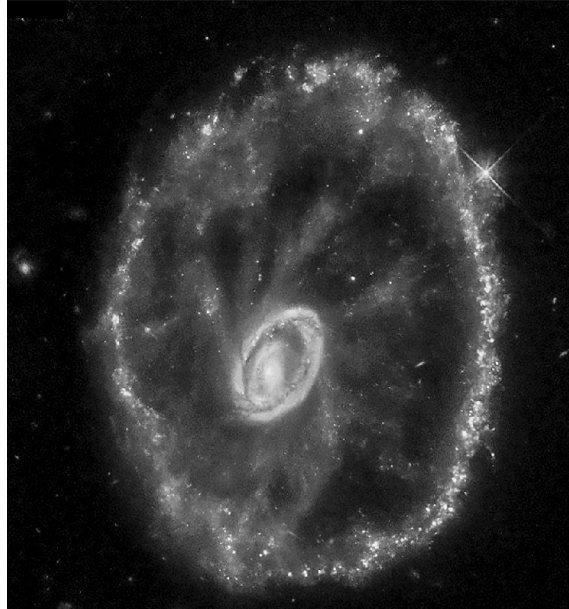


0	7
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Galaxies consist mostly of hydrogen gas.

Figure 10 shows the Cartwheel galaxy. This galaxy was recently discovered using a space telescope close to the Earth.

Figure 10



Question 7 continues on the next page

Turn over ►



07.1

The hydrogen gas in the galaxy emits light with a wavelength of 656 nm.

Determine the frequency of the light.

$$1 \text{ nm} = 1 \times 10^{-9} \text{ m}$$

$$\text{speed of light} = 3.0 \times 10^8 \text{ m/s}$$

Give your answer to 2 significant figures.

[4 marks]

Frequency (2 significant figures) = _____ Hz



The Cartwheel galaxy is a large distance from the Earth.

The wavelength of the light detected by the space telescope from this galaxy is 682 nm.

Explain how the difference between the two wavelengths provides evidence that supports the Big Bang theory.

Do **not** include a calculation in your answer.

[6 marks]

[illegible]

10

END OF QUESTIONS



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